

Kazakh-British Technical University

7M06106 "Software Engineering"

DEVELOPMENT OF A NEWS CONTENT CLASSIFICATION SYSTEM

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Relevance of the Research

Accurate categorization is key for engaging users and navigating online content effectively



Improved discoverability



Boosted reader engagement



**Customized content
suggestions**

Research Goals

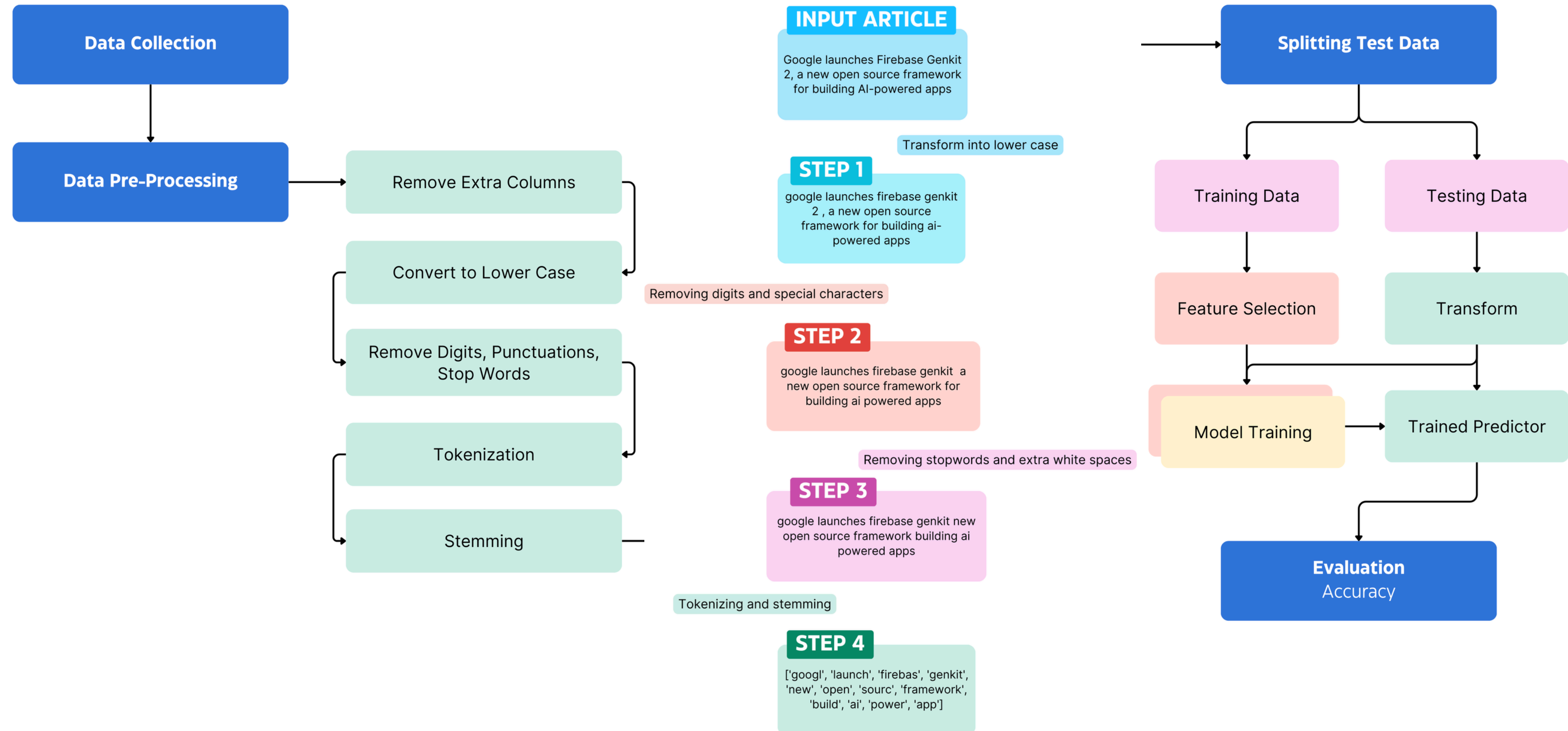
1. Develop a system for the classification of online news articles based on their content.
2. Explore the effectiveness of classical machine learning algorithms in categorizing news into predefined topics.
3. Compare multiple models to determine the most suitable algorithm for news categorization.
4. Improve accuracy, efficiency, and scalability of content classification in large datasets.



Original Dataset

ID	TITLE	URL	PUBLISHER	CATEGORY	STORY	HOSTNAME	TIMESTAMP
1	Fed official says weak data caused by weather, should r	http://www.latimes.com/business/money/la-fi-mo	Los Angeles Times	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.latimes.com	1,3945E+12
2	Fed's Charles Plosser sees high bar for change in pace	http://www.livemint.com/Politics/H2EwJSK2VE6C	Livemint	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.livemint.com	1,3945E+12
3	US open: Stocks fall after Fed official hints at accelerate	http://www.ifamagazine.com/news/us-open-stock	IFA Magazine	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.ifamagazine.com	1,3945E+12
4	Fed risks falling 'behind the curve', Charles Plosser says	http://www.ifamagazine.com/news/fed-risks-fallin	IFA Magazine	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.ifamagazine.com	1,3945E+12
5	Fed's Plosser: Nasty Weather Has Curbed Job Growth	http://www.moneynews.com/Economy/federal-res	Moneynews	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.moneynews.com	1,3945E+12
6	Plosser: Fed May Have to Accelerate Tapering Pace	http://www.nasdaq.com/article/plosser-fed-may-f	NASDAQ	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.nasdaq.com	1,3945E+12
7	Fed's Plosser: Taper pace may be too slow	http://www.marketwatch.com/story/feds-plosser-	MarketWatch	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.marketwatch.com	1,3945E+12
8	Fed's Plosser expects US unemployment to fall to 6.2%	http://www.fxstreet.com/news/forex-news/article	FXstreet.com	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	www.fxstreet.com	1,3945E+12
9	US jobs growth last month hit by weather:Fed President	http://economictimes.indiatimes.com/news/interr	Economic Times	b	ddUyU0VZz0BRneMioxUPQVP6slxvM	economictimes.indiatimes.com	1,3945E+12
10	ECB unlikely to end sterilisation of SMP purchases - trad	http://www.iii.co.uk/news-opinion/reuters/news/1	Interactive Investor	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.iii.co.uk	1,3945E+12
11	ECB unlikely to end sterilization of SMP purchases: trad	http://in.reuters.com/article/2014/03/10/us-ecb-r	Reuters India	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	in.reuters.com	1,3945E+12
12	EU's half-baked bank union could work	http://blogs.reuters.com/hugo-dixon/2014/03/10/e	Reuters UK \ (blog\)	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	blogs.reuters.com	1,3945E+12
13	Europe reaches crunch point on banking union	http://in.reuters.com/article/2014/03/10/eu-banks	Reuters	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	in.reuters.com	1,3945E+12
14	ECB FOCUS-Stronger euro drowns out ECB's message t	http://in.reuters.com/article/2014/03/10/ecb-polit	Reuters	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	in.reuters.com	1,3945E+12
15	EU aims for deal on tackling failing banks	http://main.omanobserver.om/\?p=63376	Oman Daily Observer	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	main.omanobserver.om	1,3945E+12
16	Forex - Pound drops to one-month lows against euro	http://www.nasdaq.com/article/forex-pound-drop	NASDAQ	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.nasdaq.com	1,3945E+12
17	Noyer Says Strong Euro Creates Unwarranted Economic	http://www.sfgate.com/business/bloomberg/articl	San Francisco Chronicle	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.sfgate.com	1,3945E+12
18	EU Week Ahead March 10-14: Bank Resolution, Transpa	http://blogs.wsj.com/brussels/2014/03/10/eu-wee	Wall Street Journal \ (blog\)	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	blogs.wsj.com	1,3945E+12
19	ECB member Noyer is 'very open to all kinds of tools'	http://www.ifamagazine.com/news/ecb-member-	IFA Magazine	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.ifamagazine.com	1,3945E+12
20	Euro Anxieties Wane as Bunds Top Treasuries, Spain De	http://www.businessweek.com/news/2014-03-10/	Businessweek	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.businessweek.com	1,3945E+12
21	Noyer Says Strong Euro Creates Unwarranted Economic	http://www.businessweek.com/news/2014-03-10/	Businessweek	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.businessweek.com	1,3945E+12
22	Noyer Says Stronger Euro Creates Unwarranted Pressu	http://www.sfgate.com/business/bloomberg/articl	San Francisco Chronicle	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.sfgate.com	1,3945E+12
23	Bad loan triggers key feature in ECB bank test announce	http://www.globalpost.com/dispatch/news/thoms	GlobalPost	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.globalpost.com	1,3945E+12
24	China's trade deficit and structural worries in Europe, a	http://www.fxstreet.com/analysis/strategic-curren	FXstreet.com	b	dPhGU51DcrolUIMxbRm0InaHGA2XM	www.fxstreet.com	1,3945E+12

News article classification process



Methods Details

TF-IDF to convert text data into numerical

Naive Bayes algorithm for classification

```
# Feature Engineering: using TF-IDF
tfidf_vectorizer = TfidfVectorizer()
X = tfidf_vectorizer.fit_transform(df['clean_title'])
y = df['CATEGORY']

# Model Training: Train a Naive Bayes classifier
X_train, X_test, y_train, y_test = \
    train_test_split(X, y, test_size=0.2, random_state=42)
classifier = MultinomialNB()
classifier.fit(X_train, y_train)

# Model Evaluation
y_pred = classifier.predict(X_test)
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Theoretical and Practical Significance

1. Helps developers choose the right algorithm for similar tasks
2. Demonstrates effective preprocessing techniques for NLP
3. Can be integrated into media and content platforms

Theoretical and Practical Novelty

1. Comparative analysis of multiple models on the same dataset
2. Application of standard NLP techniques on real-world data
3. Use of classic ML instead of deep learning for efficient classification

Overview of Current Research in the Field

Chen, Z., Ni, C., & Murphey, Y. L. (2006)

Neural Network Approaches for Text Document Categorization. IEEE IJCNN.

Zhang, X., Li, B., & Sun, X. (2010)

A K-Nearest Neighbor Text Classification Algorithm Based on Fuzzy Integral. ICNC 2010.

Bijalwan, V., Kumar, V., Kumari, P., & Pascual, J. (2014)

KNN-Based Machine Learning Approach for Text and Document Mining. IJDTA, 7(1).

This study introduced hierarchical and categorical neural networks for document classification. It showed that combining multiple networks improves accuracy, especially in high-dimensional feature spaces. However, the focus was limited to engineering diagnostic documents.

Proposed a fuzzy integral-based method to improve KNN for text classification. While effective, it mainly targets the integration of evidence from neighboring samples and lacks comparison with modern ensemble models.

This work applied KNN for document categorization and retrieval, focusing on simplicity and relevance. It provided a baseline method but did not explore hybrid or deep learning techniques.

Results

Classification Report:

	precision	recall	f1-score	support
b	0.89	0.91	0.90	23414
e	0.95	0.97	0.96	30353
m	0.97	0.84	0.90	9024
t	0.90	0.90	0.90	21693
accuracy			0.92	84484
macro avg	0.93	0.91	0.92	84484
weighted avg	0.92	0.92	0.92	84484

Model	Accuracy
SVM	0.9206
LR	0.8560
KNN	0.8431

Future Works

Exploration of Advanced Algorithms

Investigate Ensemble Methods

Deep Learning Models

Conclusion

Accomplished a sorting accuracy of 92% with news articles.

The models performed well on different data conditions and forms such as noise, imbalance, and feature representation

Publications by the Master's Student

Universum:
Technical Sciences

- Title: DEVELOPMENT OF AN AI-BASED SYSTEM FOR GENERATING MARKETING CREATIVES FOR CAMPAIGNS AND PROMOTIONS

СПРАВКА

26.05.2025 г. № 36236

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Ziro Aasso Araz

Редакция журнала ««Universum: технические науки»» сообщает Вам о том, что Ваша статья «DEVELOPMENT OF AN AI-BASED SYSTEM FOR GENERATING MARKETING CREATIVES FOR CAMPAIGNS AND PROMOTIONS» принята к публикации в выпуске 6(135), который будет опубликован на сайте <https://7universum.com/ru/tech> 28 июня 2025 года.

Руководитель проекта
Научные журналы «Universum»
Генеральный директор
Издательства «МЦНО»



И.С. Степанов



Publications by the Master's Student

Eurasian Journal of Mathematical and Computer Applications
(Scopus Q3) – Under Review

- Title: ***A Hybrid Agile Approach for Developing a Web-Based Classification System in a News Organization***



EURASIAN JOURNAL OF MATHEMATICAL AND COMPUTER APPLICATIONS

Dear Authors,

Thank you for your submission to the *Eurasian Journal of Mathematical and Computer Applications*.

Authors: Akhmadi Nurbol, Makulbay Azamat

Manuscript Title: *A Hybrid Agile Approach for Developing a Web-Based Classification System in a News Organization*

Date of Submission: April 3, 2025

We hereby acknowledge receipt of your manuscript titled "*A Hybrid Agile Approach for Developing a Web-Based Classification System in a News Organization*." Please be advised that it is currently under review by the editorial board.

Should any additional information be required during the review process, we will contact you accordingly.

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Thank you!

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